

Rates-Motion-Time-and-Geometry

CSAT semantic master • csat-04 • g1 • approved: false

01

Rate model

Quantity equals rate multiplied by time.

02

Time and work

Add work rates, not completion times.

03

Pipes and leaks

Treat filling positive and emptying negative.

04

Speed and distance

Match units before using distance equals speed times time.

05

Relative motion

Add opposite-direction speeds and subtract same-direction speeds.

06

Trains and boats

Include object length and current speed correctly.

07

Races and tracks

Use relative distance on linear or circular paths.

08

Clocks

Use the 5.5-degree-per-minute relative hand speed.

09

Calendars

Reduce day shifts modulo seven and handle leap-year rules.

10

Geometry

Use properties before coordinates or formulas.

11

Mensuration

Distinguish length, area and volume units.

12

Verification

Check dimensions, magnitude, endpoints and physical possibility.